

CHAPTER 16

SOUND

3.4 Sound

- 1 Describe the production of sound by vibrating sources
- 2 Describe the longitudinal nature of sound waves and describe compressions and rarefactions
- 3 State the approximate range of frequencies audible to humans as 20 Hz to 20 000 Hz
- 4 Explain why sound waves cannot travel in a vacuum and describe an experiment to demonstrate this
- 5 Describe how changes in amplitude and frequency affect the loudness and pitch of sound waves
- 6 Describe how different sound sources produce sound waves with different qualities (timbres), as shown by the shape of the traces on an oscilloscope
- 7 Describe an echo as the reflection of sound waves
- 8 Describe simple experiments to show the reflection of sound waves
- 9 Describe a method involving a measurement of distance and time for determining the speed of sound in air
- 10 Know that the speed of sound in air is approximately 330–350 m/s
- 11 Know that, in general, sound travels faster in solids than in liquids and faster in liquids than in gases
- 12 Define ultrasound as sound with a frequency higher than 20 kHz
- 13 Describe the uses of ultrasound in cleaning, prenatal and other medical scanning, and in sonar (including calculation of depth or distance from time and wave speed)

1. O/N/23/11/Q22

Which range of sound frequencies includes only frequencies of sound that can be heard by a healthy human ear?

- A** 0.5–50 Hz **B** 5–500 Hz **C** 50–5000 Hz **D** 500–50 000 Hz

2. M/J/23/12/Q23

Ultrasound is a type of sound wave that is not audible to a healthy human ear.

Which range of frequencies contains **only** ultrasound?

- A** 30 Hz–3.0 kHz
B 300 Hz–30 kHz
C 3.0 kHz–3.0 MHz
D 30 kHz–30 MHz

3. M/J/23/11/Q24

Sound travels at different speeds in different states of matter.

Which list gives the three states of matter arranged in order of the speed of sound from slowest to fastest?

- A** gas → liquid → solid
B liquid → solid → gas
C solid → gas → liquid
D solid → liquid → gas

4. O/N/22/11/Q25

An oscilloscope is used to demonstrate a sound.

Which feature of the oscilloscope trace is affected by the quality (timbre) of the sound?

- A** amplitude
B frequency
C shape
D wavelength

5. O/N/22/12/Q27

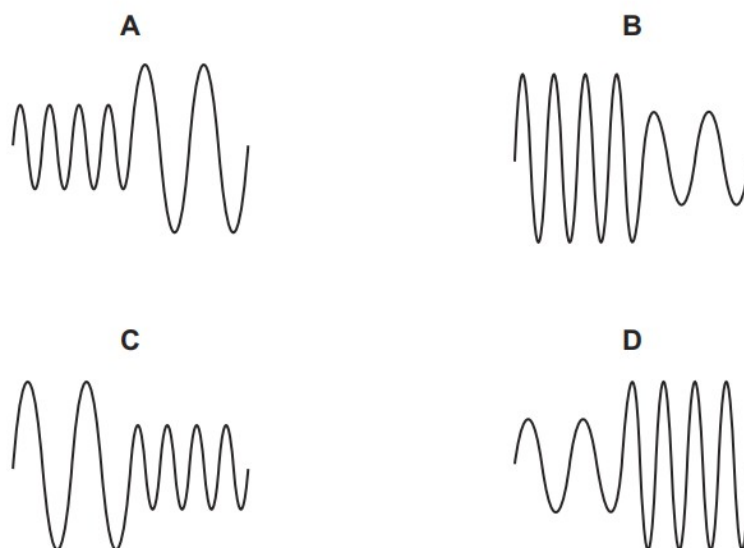
Which two frequencies are both outside the range of audible frequencies for a healthy human ear?

- A** 2.0 Hz and 500 Hz
B 2.0 Hz and 25 kHz
C 25 Hz and 25 kHz
D 500 Hz and 25 kHz

6. O/N/22/12/Q28

Sound waves are displayed as a trace on the screen of an oscilloscope.

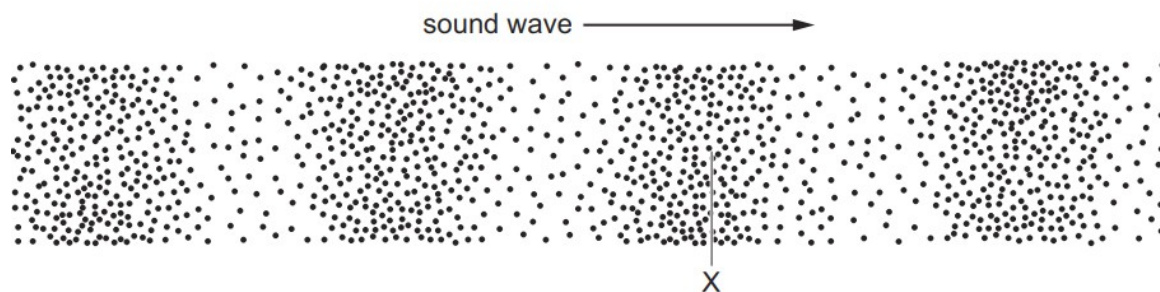
Which trace shows a sound that becomes quieter with a higher pitch?



7. M/J/22/12/Q27

A sound wave passes through a substance.

The diagram shows the position of molecules of the substance at one point in time.



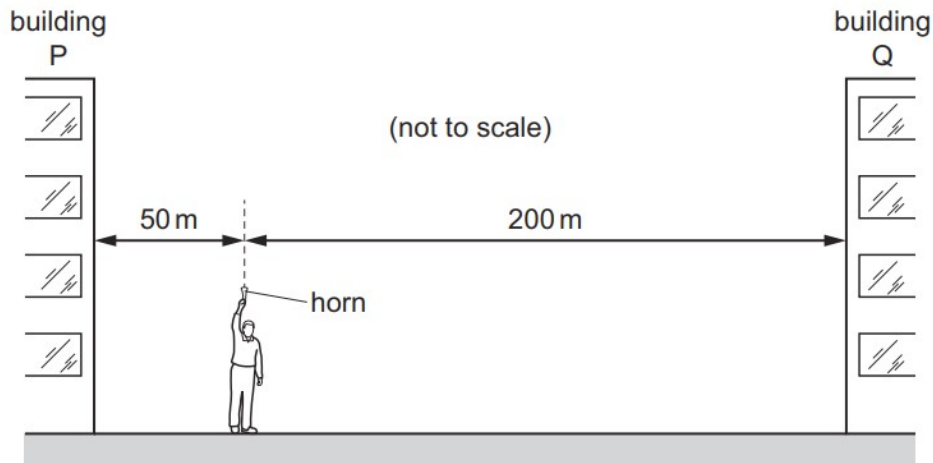
Which type of wave is a sound wave and what is at X?

	type of wave	at X
A	longitudinal	compression
B	longitudinal	rarefaction
C	transverse	compression
D	transverse	rarefaction

8. M/J/22/12/Q28

A man stands between two tall buildings, P and Q.

He is 50 m from P and 200 m from Q.



He sounds a horn.

He hears the first echo from building P and 1.0 s later he hears the first echo from building Q.

What is the speed of sound calculated using this information?

- A** 250 m/s **B** 300 m/s **C** 330 m/s **D** 500 m/s

9. M/J/22/12/Q29

What is one of the uses of ultrasound?

- A** cleaning jewellery
B fluorescent tubes
C optical fibres
D sunbeds

10. M/J/22/11/Q25

Sound travels at different speeds in each of the three states of matter, solid, liquid and gas.

Which row shows where sound travels fastest and where sound travels slowest?

	state where sound travels fastest	state where sound travels slowest
A	gas	liquid
B	gas	solid
C	liquid	gas
D	solid	gas

11. O/N/21/11/Q25

A student stands a distance x in front of a large wall.

He claps his hands at a regular rate so that each clap coincides with the echo from the previous clap.

In a time t , he claps his hands N times.

Which expression is used to calculate the speed of sound in air?

A $\frac{x}{Nt}$

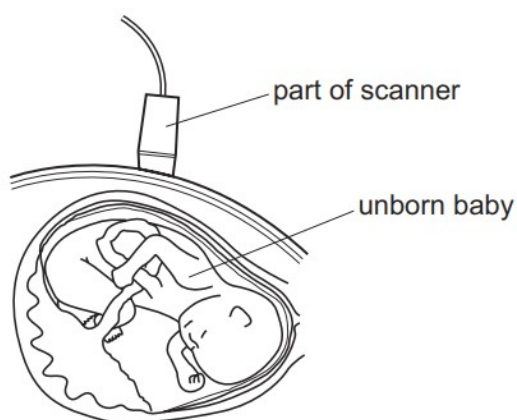
B $\frac{2x}{Nt}$

C $\frac{Nx}{t}$

D $\frac{2Nx}{t}$

12. O/N/21/11/Q26

An ultrasound scanner produces an image of an unborn baby.



What does the scanner use to form an image of the baby?

A ultrasound absorbed by the baby

B ultrasound emitted by the baby

C ultrasound reflected by the baby

D ultrasound refracted by the baby

13. O/N/21/12/Q28

Which of the following is most closely associated with echoes?

A focussing

B reflection

C refraction

D total internal reflection

14. M/J/21/12/Q28

The sound from a ship is reflected by a cliff. An echo is heard by a sailor on the ship 4.0 s after the sound is made. The speed of sound in air is 320 m/s.

How far from the cliff is the ship?

- A** 160 m **B** 640 m **C** 1280 m **D** 2560 m

15. M/J/21/12/Q29

Students are asked for the uses of ultrasound.

Three suggested uses for ultrasound are:

- P** cleaning jewellery
- Q** tanning on a sunbed
- R** obtaining an image of an unborn baby.

Which suggestions are correct?

- A** P and Q only **B** P and R only **C** Q and R only **D** P, Q and R

16. O/N/20/11/Q30

Which description of a sound wave is correct?

- A** a longitudinal electromagnetic wave
- B** a transverse electromagnetic wave
- C** an oscillation of particles parallel to the direction of travel of the wave energy
- D** an oscillation of particles perpendicular to the direction of travel of the wave energy

17. O/N/20/11/Q31

Where does sound travel the fastest?

- A** in a vacuum
- B** in gases
- C** in liquids
- D** in solids

18. O/N/20/12/Q27

A scientific research boat uses pulses of ultrasound to detect fish underneath it.

A pulse of ultrasound is bounced from a group of fish 6000 m directly below the boat.

The next pulse is sent out 2.0 s after the echo from the previous pulse has been received.

The speed of ultrasound in water is 1500 m/s.

What is the interval between the two pulses?

- A** 4.0 s **B** 6.0 s **C** 8.0 s **D** 10 s

19. O/N/20/12/Q28

Which row correctly describes ultrasound?

	type of wave	frequency
A	longitudinal	less than 20 kHz
B	longitudinal	greater than 20 kHz
C	transverse	less than 20 kHz
D	transverse	greater than 20 kHz

20. M/J/20/11/Q30

What is the ratio of the longest sound wavelength audible to a healthy human ear to the shortest?

- A** 20:1 **B** 1000:1 **C** 10 000:1 **D** 20 000:1

21. M/J/20/11/Q31

Ultrasound is used to clean jewellery in a liquid.

What is another use of ultrasound?

- A** optical fibre communication
B pre-natal scanning
C sunbeds
D telephone communications

22. O/N/19/11/Q28

A person hears an echo after a sound hits a solid cliff face.

What causes the echo?

- A** absorption
B dispersion
C reflection
D refraction

23. O/N/19/11/Q29

What is a typical value for the speed of sound in air?

- A** 3.3×10^4 cm/s
B 3.3×10^4 m/s
C 3.3×10^2 km/s
D 3.3×10^8 m/s

24. O/N/19/12/Q27

Which row shows the speed of sound in air (c_a), liquid (c_L) and solid (c_s) in order from slowest to fastest?

	slowest	→	fastest
A	c_L		c_s
B	c_L		c_a
C	c_a		c_s
D	c_s		c_L

25. O/N/19/12/Q28

A man stands 80 m in front of a cliff face. He makes a loud bang and listens for the echo.

He makes a loud bang once every second. He hears an echo exactly halfway between the bang that caused it and the next bang.

What is the speed of sound?

- A** 40 m/s **B** 80 m/s **C** 160 m/s **D** 320 m/s

26. M/J/19/12/Q29

What is the approximate range of audible frequencies for a healthy human ear?

- A** 2.0 Hz to 2.0 kHz
B 20 Hz to 2.0 kHz
C 20 Hz to 20 kHz
D 200 Hz to 200 kHz

27. M/J/19/12/Q30

The speed of sound in air is c_a , the speed of sound in water is c_w and the speed of sound in ice is c_i .

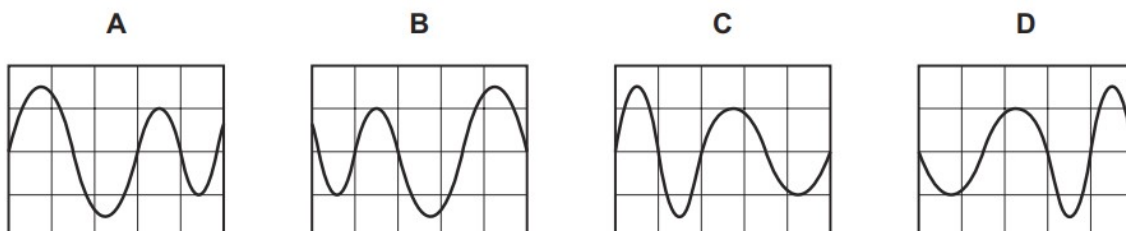
Which relationship is correct?

- A** $c_a > c_i$ **B** $c_a > c_w$ **C** $c_i > c_w$ **D** $c_w = c_a$

28. M/J/19/11/Q27

Four sound waves are displayed on the screen of a cathode-ray oscilloscope.

Which sound wave gets louder and has a pitch that decreases?



29. O/N/18/11/Q21

What is the approximate value for

$$\frac{\text{speed of light in air}}{\text{speed of sound in air}} ?$$

- A** 10^2 **B** 10^4 **C** 10^6 **D** 10^8

30. O/N/18/11/Q22

What are the lowest frequency and the highest frequency of the sound that a human being with normal hearing can hear?

	lowest frequency	highest frequency
A	2.0 Hz	2.0 kHz
B	2.0 Hz	20 000 Hz
C	20 Hz	20 kHz
D	20 kHz	20 000 kHz

31. O/N/18/11/Q23

Ultrasound has a number of uses in engineering and medicine.

What is ultrasound **not** used for?

- A** cleaning
B pre-natal scanning
C quality control
D sterilising medical equipment

32. O/N/18/12/Q27

Ultrasound is used to map the ocean floor.

During one survey, the depth of water is 1200 m. An ultrasound pulse is sent from the surface and when it returns to the ship, another pulse is sent immediately. In any period of 8.0 s, five pulses are sent down from the surface.

What is the speed of the ultrasound in water?

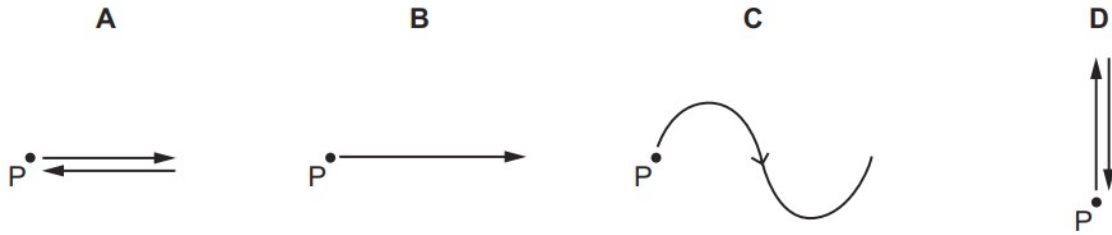
- A** 150 m/s **B** 300 m/s **C** 750 m/s **D** 1500 m/s

33. M/J/18/12/Q31

The diagram shows a loudspeaker that is producing a continuous sound wave of frequency 200 Hz in air.



Which diagram best shows how the sound wave causes a molecule at P to move during $\frac{1}{200}$ s ?



34. M/J/18/12/32

Which frequency is in the range heard by a healthy human ear?

- A** 12 mHz **B** 12 Hz **C** 12 kHz **D** 12 MHz

35. M/J/18/12/Q33

What is one of the uses of ultrasound?

- A** cleaning jewellery
- B** satellite communication
- C** fluorescent tubes
- D** optical fibres

36. M/J/18/11/Q25

A young person with healthy ears can hear a range of frequencies.

What is the approximate range of frequencies?

- A** 2 Hz to 2000 Hz only
- B** 2 Hz to 20 000 Hz only
- C** 20 Hz to 2000 Hz only
- D** 20 Hz to 20 000 Hz only

37. O/N/17/11/Q24

Which row gives the speed of sound in air, in water and in steel?

	<u>speed in air</u> m/s	<u>speed in water</u> m/s	<u>speed in steel</u> m/s
A	330	1500	6000
B	330	6000	1500
C	6000	330	1500
D	6000	1500	330

38. O/N/17/11/Q25

As a sound wave travels from one medium to another, its wavelength increases.

What happens to the frequency and to the speed of the sound?

	frequency	speed
A	decreases	decreases
B	decreases	increases
C	stays constant	decreases
D	stays constant	increases

39. M/J/17/11/Q31

A ship that is stationary on the surface of the sea sends pulses of sound vertically downwards towards the sea bed.

Each pulse that reflects from the sea bed is received 1.0 s after it is sent out.

A whale swims under the boat and a pulse is received 0.60 s after it is sent out.

The speed of sound in sea water is 1500 m/s.

What is the distance of the whale above the sea bed?

- A** 300 m **B** 450 m **C** 600 m **D** 750 m

40. O/N/16/11/Q29

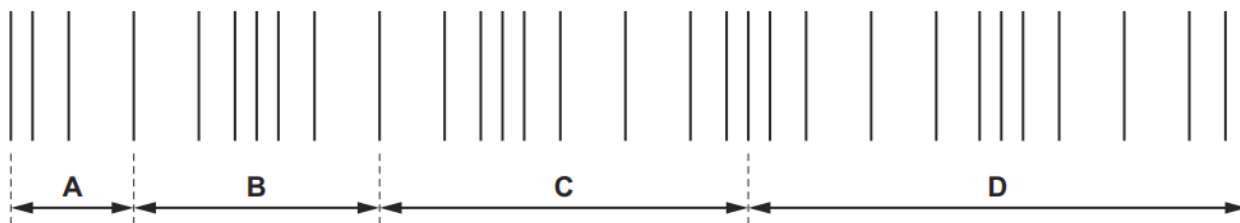
Which property of a sound wave increases as the loudness of the sound increases?

- A** amplitude
B frequency
C speed
D wavelength

41. M/J/16/12/Q22

A sound wave travels through air. The lines in the diagram show the positions of layers of air at one particular time.

Which distance shows the wavelength of the wave?



42. M/J/16/11/Q26

An intruder alarm is adjusted to give a quieter sound without affecting the pitch of the note.

How are the amplitude and the frequency of the sound affected?

	amplitude	frequency
A	lower	lower
B	lower	same
C	same	lower
D	same	same

43. O/N/15/11//Q22

A musical note is produced by two sources.

The traces produced by each source on the screen of a cathode-ray oscilloscope (c.r.o.) are shown below.



What is the difference between the sounds?

- A** the amplitude
- B** the frequency
- C** the pitch
- D** the quality

44. O/N/15/12/Q25

A loudspeaker produces a sound wave of frequency 50 Hz.
The amplitude of the sound wave is increased.

What is heard?

- A** a louder sound of a higher pitch
- B** a louder sound of the same pitch
- C** a sound of higher pitch but the same loudness
- D** a sound of the same pitch and loudness as before

45. O/N/15/12/Q27

The sound from the siren of a ship is reflected by a cliff. An echo is heard by a sailor on the deck of the ship, 4.0 s after the siren is sounded. The speed of sound in air is 320 m/s.

How far from the cliff is the ship?

- A** 80 m
- B** 160 m
- C** 640 m
- D** 1280 m

46. M/J/15/12/Q30

Two sound waves X and Y are compared.

X has the greater frequency.

Y has the greater amplitude.

How do the loudness and pitch of sound wave Y compare to those of X?

- A** Y is louder and higher pitch.
- B** Y is louder and lower pitch.
- C** Y is quieter and higher pitch.
- D** Y is quieter and lower pitch.

47. O/N/14/11/Q25

A star explodes in outer space.

Which waves from the exploding star do **not** reach the Earth?

- A** infra-red
- B** light
- C** radio
- D** sound

48. O/N/14/12/Q18

A student stands at a distance d from the base of a tall cliff.

He claps together two pieces of wood and measures the time that elapses before he hears the echo. He conducts the experiment five times and obtains these results.

0.72 s 0.80 s 0.71 s 0.81 s 0.71 s

The speed of sound is 320 m/s.

What is the distance d ?

- A** 120 m **B** 240 m **C** 480 m **D** 600 m

49. O/N/14/12/Q19

Ultrasound has many uses.

For what are ultrasound waves used?

- A** killing cancerous cells
B pre-natal scanning
C sunbeds
D telephones

50. M/J/14/12/Q28

Which frequency is in the ultrasound range?

- A** 35 Hz **B** 350 Hz **C** 3500 Hz **D** 35 000 Hz

51. M/J/14/12/Q29

In an experiment to determine the speed of sound in air, a student stands 200 m away from a cliff and claps two pieces of wood together.

His class-mates standing next to him start stopwatches when the two pieces of wood meet and stop the stopwatches when they hear the echo.

Their times are:

1.44 s 1.70 s 1.58 s 1.76 s

Which value for the speed of sound do they obtain?

- A** 62 m/s **B** 123 m/s **C** 247 m/s **D** 340 m/s

52. M/J/14/11/Q28

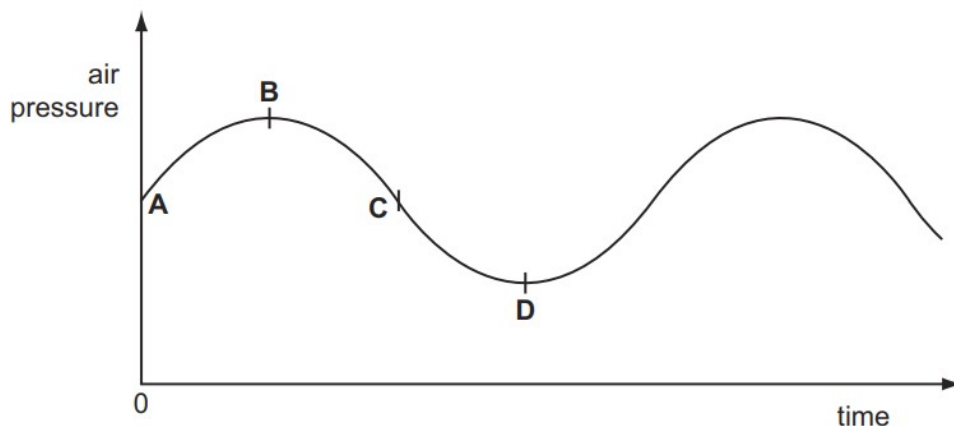
Which statement about the speed of sound is correct?

- A** sound travels fastest in a vacuum
B sound travels fastest in gases
C sound travels fastest in liquids
D sound travels fastest in solids

53. O/N/13/11/Q23

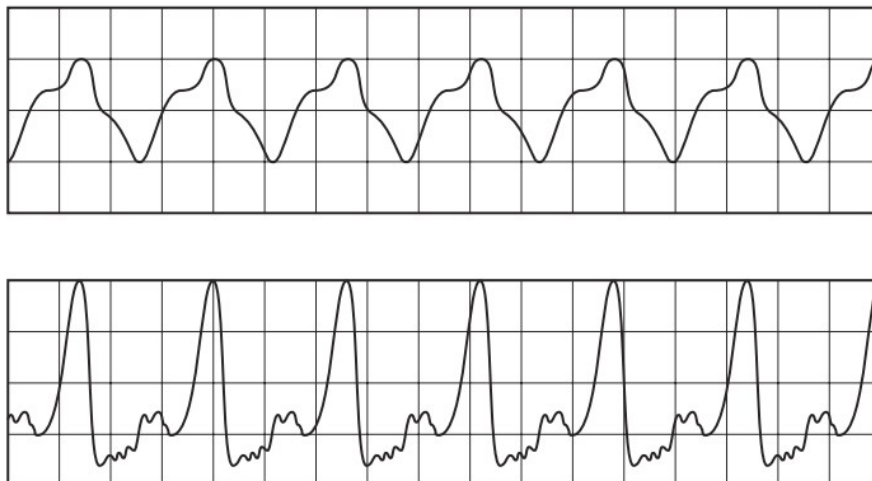
The graph shows how the pressure varies as a sound wave passes through air.

Which point represents a compression?



54. O/N/13/11/Q24

The sounds produced by two musical instruments are directed towards a microphone connected to an oscilloscope (c.r.o.). The waveforms produced on the screen are shown.



The waveforms show that the sounds produced have a different property.

What is the property?

- A** frequency
- B** speed
- C** timbre (quality)
- D** wavelength

55. O/N/13/12/Q25

Two campers are woken up in the middle of the night by a thunderstorm. Their tent is lit up by a flash of lightning and they hear the thunderclap 5.0 s later. The speed of sound is 340 m/s.

How far away from the tent is the lightning?

- A** 68 m **B** 850 m **C** 1700 m **D** 3400 m

56. M/J/13/12/Q27

What is a possible frequency of an ultrasound wave?

- A** 0.1 kHz **B** 3 kHz **C** 10 kHz **D** 30 kHz

57. M/J/13/12/Q28

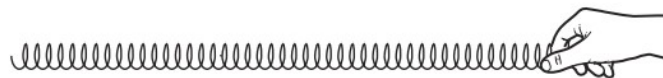
A guitar string is made to vibrate.

What makes the pitch of the note rise?

- A** a decrease in the amplitude of vibration
B a decrease in the frequency of vibration
C an increase in the amplitude of vibration
D an increase in the frequency of vibration

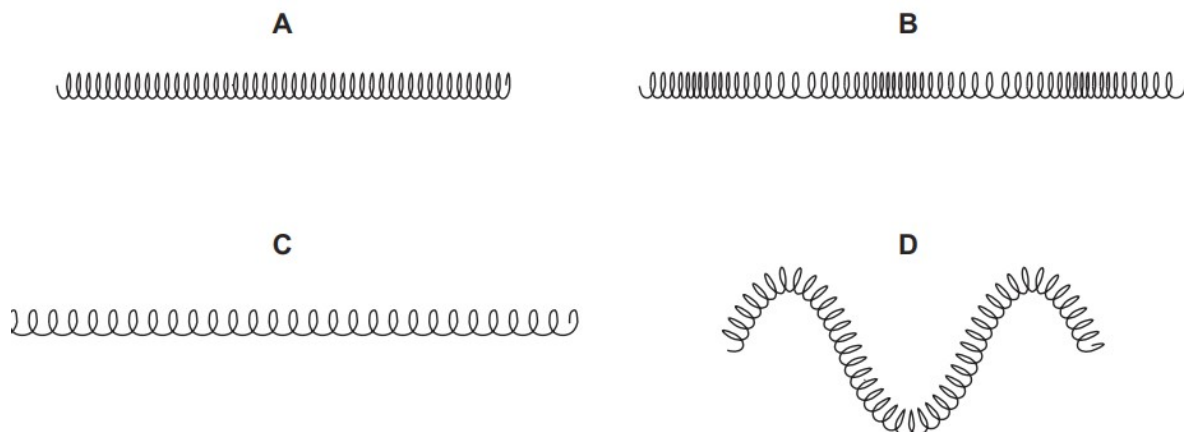
58. O/N/11/11/Q23

A student uses a spring to demonstrate waves. He moves the spring with his hand.



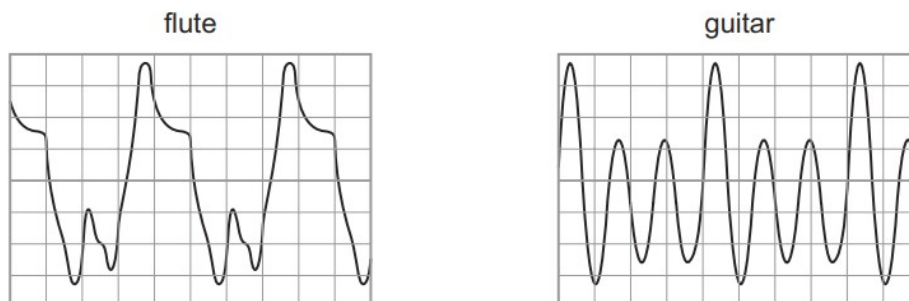
spring placed on bench

Which diagram demonstrates the type of wave produced by a source of sound?



59. O/N/11/12/Q25

Waveforms are shown on a cathode-ray oscilloscope for a flute and for a guitar, each playing the same note. The oscilloscope settings are the same for both waveforms.



What is the difference between the two sounds?

- A** the amplitude
- B** the frequency
- C** the quality (timbre)
- D** the wavelength

60. M/J/11/12/Q26

Sound travels at different speeds in air, water and steel.

For these materials, which row is correct?

	sound travels slowest in	sound travels fastest in
A	air	steel
B	air	water
C	steel	air
D	water	air

61. O/N/10/11/Q25

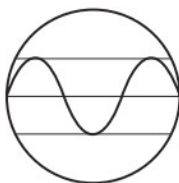
During a thunderstorm, there is an interval of 1.70 s between an observer seeing the lightning and hearing the thunder. The speed of sound is 340 m/s.

What is the distance between the observer and the storm?

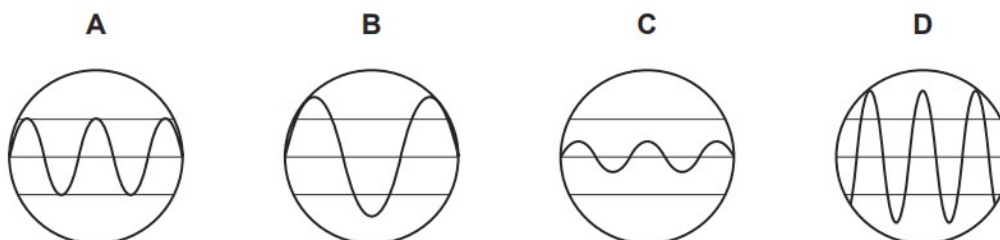
- A** 100 m **B** 200 m **C** 578 m **D** 1160 m

62. O/N/09/Q23

The diagram shows the trace produced on a cathode-ray oscilloscope (c.r.o.) by a sound.



Which trace is produced when both the loudness and the pitch of the sound are increased?



63. M/J/09/Q23

An ultrasonic tape-measure is used to find the distance to a wall. It sends out an ultrasonic pulse and times how long it takes for the reflected pulse to return from the wall.

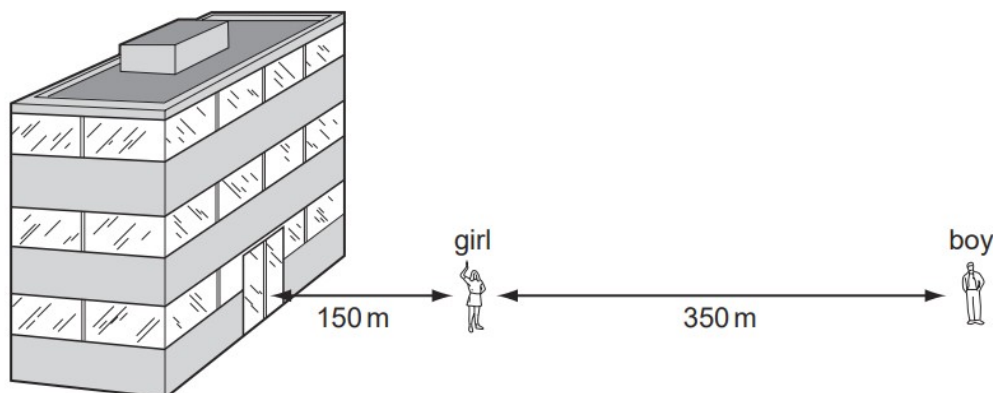
The ultrasound has a frequency, a wavelength and a speed.

Which pair of values is needed to find the distance to the wall?

- A** frequency and wavelength
- B** frequency and time taken for the pulse to return
- C** speed and time taken for the pulse to return
- D** wavelength and time taken for the pulse to return

64. M/J/07/Q23

A girl, standing 150 m in front of a tall building, fires a shot using a starting pistol. A boy, standing 350 m from the girl, hears two bangs 1 s apart.



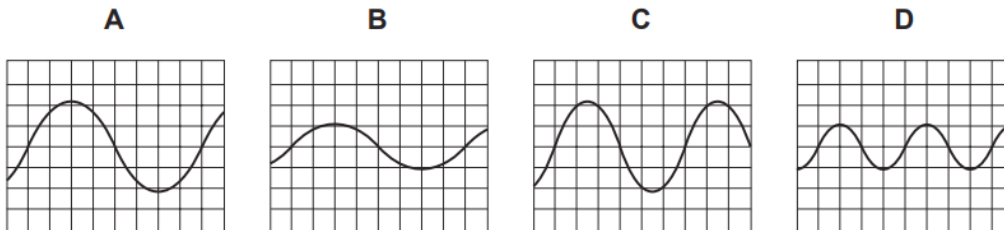
From this information, what is the speed of sound in air?

- A** 300 m/s
- B** 350 m/s
- C** 500 m/s
- D** 650 m/s

65. O/N/06/Q23

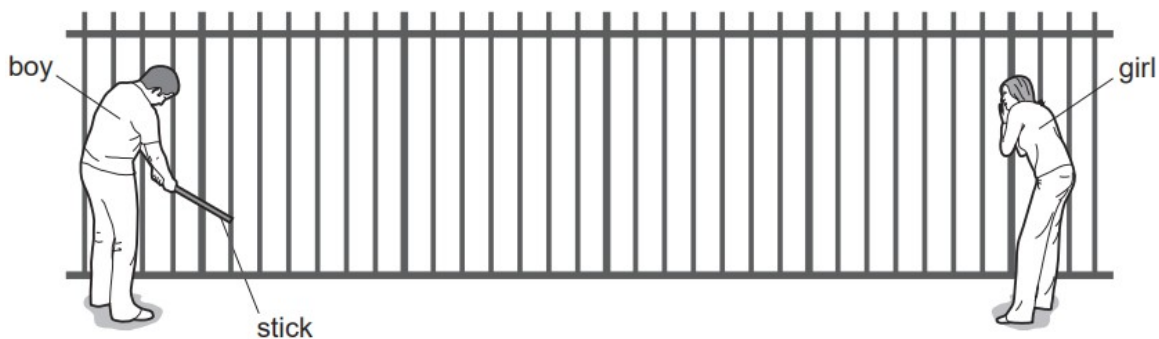
The diagrams show oscilloscope traces of sounds picked up by microphones. The oscilloscope controls are set in the same position for all the traces.

Which trace shows the sound that is both loud and low-pitched?



66. M/J/06/Q26

A boy strikes a rigid metal fence with a stick to create a sound along the fence. A girl listens with her ear against the fence. One second after the fence is struck, the girl hears a sound through the air.

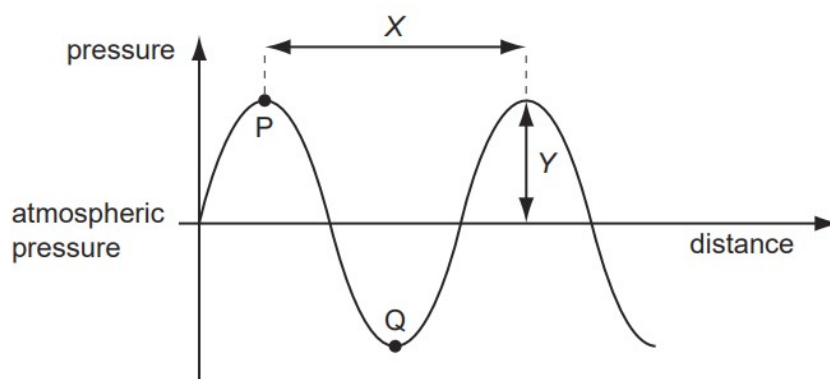


How long will it take for the sound to reach the girl through the fence?

- A** 0 second
- B** less than 1 second
- C** 1 second
- D** more than 1 second

67. M/J/05/Q24

The graph shows, at one instant, the pressure variation along a sound wave.



Which point on the diagram represents a rarefaction and what is the wavelength of the sound wave?

	rarefaction at	wavelength is
A	P	X
B	P	Y
C	Q	X
D	Q	Y